

# AgentUAV: Multi-Agent Query-Guided UAV Target Detection with Super-Resolution and SAHI

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**Abstract** — *The applications of UAV-based surveillance, search, and rescue, and aerial monitoring all demand the robust detection of targets that may be difficult to recognize due to their small size long distance partial visibility, or low image quality, among others. While conventional object detection models are mostly dependent on the predefined object categories, they may run into challenges when the target or scene category varies. Open-vocabulary detection can recognize the category of targets using the query text, but applying the same processing approach to all UAV images can mean unnecessary cost of computation or underprocessing of complex targets. To overcome these deficiencies, we propose AgentUAV—a multi-agent query-guided UAV target detection system, which combines open-vocabulary detection with Slicing Aided Hyper Inference (SAHI), Super-Resolution, and adaptive processing. The system takes as input a UAV image or video frame and a natural-language target query and endeavors to first identify the requested target with YOLO-World. The detected results are then analyzed based on target types and visual contexts to determine whether further processing is needed. Use of the SAHI based sliced inference is extended for small far away targets, whereas Super-Resolution enhancement is used to address low resolution or degraded parts of the image. Multiple adaptive specialized agents work together for query understanding, target detection, analysis, and verification, and adaptive decision-making to produce the final result of target detection. The comprehensive system is evaluated in multiple detection set-ups to investigate the advantage of sliced inference, image enhancement and adaptive multi-agent processing. The results show that combining these approaches can positively benefit the target detection coverage in challenging UAV scenarios in particular for small and hard to detect targets. The proposed AgentUAV setup brings a highly flexible approach for query-based UAV target detection, by selecting different approaches per the input image and detected target types.*

**Keywords** — UAV Target Detection, Multi-Agent Framework, Query-Guided Detection, Open-Vocabulary Object Detection, SAHI-Based Sliced Inference, Super-Resolution, Adaptive Target Detection.

## I. INTRODUCTION

The capabilities of Unmanned Aerial Vehicles (UAVs) in such areas as surveillance, target detection, environmental monitoring and search and rescue have been broadened in recent years by the likes of artificial intelligence, computer vision and deep learning. Yet there are limitations; UAV imagery is replete with targets that are small, far off, of low

resolution or only partially in view, which defies conventional detection. While open-vocabulary models like YOLO-World can be used to zero in on a target via natural-language queries, one cannot count on a single method to be dependable in all aerial scenarios. And to put every image through complex processing is to invite higher computational costs.

There is a case for an adaptive approach. SAHI, for instance, handles small objects well by breaking down large images into overlapping pieces so that a distant target has more effective area at inference time. Super-Resolution can do the same for poor quality regions. But not every UAV image calls for this. An adaptive system would be better served to look at the initial results and decide if further processing is warranted on the basis of the target's size, distance and the overall image quality.

To that end, we present AgentUAV, a Multi-Agent Query-Guided framework for UAV target detection. It is a system that integrates YOLO-World with SAHI sliced inference, Super-Resolution and adaptive decision-making. Upon receiving a frame from a UAV along with a natural-language query, the system will carry out an initial pass and assess the conditions of what it has found. Should the analysis indicate a need, it will apply SAHI or Super-Resolution before verifying the detection and producing the final output. We have put the framework to the test with UAV aerial imagery, measuring Precision, Recall, mAP@50 and F1-Score to see how it holds up in difficult detection environments.

## II. LITERATURE SURVEY

Recent advances in UAV-based object detection and open-vocabulary detection contribute to the construction of more efficient systems for aerial target detection. Cheng et al. [1] developed YOLO-World, an open-vocabulary object detector, capable of real-time detection based on natural language inquiries instead of standard predefined categories. On the other hand, Akyon et al. [2] addressed the problem of small-object detection in computer vision tasks via developing Slicing Aided Hyper Inference (SAHI), a sliced image inference tool that divides images into smaller regions to improve small and distant object detection. Wang et al. [3] introduced Real-ESRGAN for real-world Super-Resolution applications which can reconstruct visually appealing results from significantly degraded images. Thus, the aforementioned works highlight the importance of employing more flexible target recognition models combined with additional processing steps for detecting challenging scenes in UAV images/videos.

Traditional object detection architectures have also played an important role in the development of efficient detection systems. Redmon et al. [7] introduced YOLO as a unified real-time

object detection architecture, allowing objects to be detected efficiently within a single framework. Redmon and Farhadi [8] later presented YOLOv3 as an improvement to the original YOLO architecture. Similarly, Wang et al. [9] proposed ESRGAN for enhanced image Super-Resolution, while Real-ESRGAN [3] further addressed Super-Resolution for real-world degraded images. Zhu et al. [10] presented the VisDrone-DET benchmark, which highlights the challenges involved in detecting objects with different scales, densities, and appearances in drone imagery. These studies have established important foundations for aerial object detection, small-object analysis, and image enhancement. However, the individual techniques generally focus on a specific processing task rather than dynamically selecting the appropriate method according to the characteristics of each UAV scene.

Existing research has therefore made considerable progress in open-vocabulary detection, small-object detection, aerial image analysis, and Super-Resolution. However, many existing approaches continue to treat detection, sliced inference, and image enhancement as separate processing stages. A fixed detection pipeline may apply the same processing strategy to every input, even when some UAV images can be processed directly while others contain small, distant, partially visible, or low-resolution targets that require additional processing. This creates a need for an adaptive framework that can analyze the initial detection results and determine whether further processing is required. In addition, combining natural-language target queries with adaptive processing can provide greater flexibility for UAV surveillance and target-search applications.

We present here AgentUAV, a multi-agent query-guided UAV target detection framework that integrates YOLO-World-based open-vocabulary detection [1], SAHI-based sliced inference [2], and Real-ESRGAN Super-Resolution enhancement [3]. The framework initially processes the UAV image or video frame according to the user-provided target query and then analyzes the obtained detection results. Based on the target characteristics and visual conditions, the system can selectively apply sliced inference or image enhancement when required. The framework further incorporates detection verification and adaptive decision-making to coordinate the different processing stages. This approach provides a unified and flexible detection pipeline for identifying challenging targets in UAV imagery while avoiding unnecessary additional processing for relatively simple detection cases.

### III. DATASET DESCRIPTION

The proposed AgentUAV framework aims to leverage UAV and aerial imagery for the evaluation of query-driven target detection in various visual conditions. The dataset comprises aerial images with varying appearances of the target at different scales, distances, orientations, and background conditions. The benchmark is prepared to facilitate the evaluation of target detection in the context of challenging UAV scenarios, particularly for small, distant, and low-resolution targets. Public UAV datasets such as VisDrone and UAVDT can be leveraged as sources for the evaluation of the proposed detection framework. The images and corresponding annotations are subjected to preliminary inspection, and samples such as corrupted images, duplicate images, invalid annotations, and unsupported object categories are removed. The remaining images are organized according to the required detection classes and experimental settings, enabling the

consistent use of the prepared dataset during the evaluation of the AgentUAV pipeline.

The processed UAV dataset comprises standardized annotations indicating the target class and corresponding bounding-boxes in normalized form, which facilitates consistency in the alignment of aerial images, annotations, and the YOLO-World detection pipeline. The final benchmark comprises the aerial images partitioned in accordance with the project's dataset configuration into training and held-out evaluation portions. The training portion is utilized for the preparation of the detection pipeline, whereas the held-out images are reserved for the evaluation of the performance of the proposed framework. Prior to the evaluation, the dataset undergoes preliminary inspection for duplicate samples, corrupted images, invalid annotations, and inconsistencies between image files and the corresponding annotations. This preprocessing facilitates the maintenance of the quality of the benchmark and enables the evaluation to be carried out using images that were not included in the training portion. The prepared dataset can then be utilized for the comparison of the baseline YOLO-World detector, the YOLO-World with SAHI configuration, and the complete AgentUAV multi-agent framework under the same evaluation conditions.

## IV. METHODOLOGY

The proposed AgentUAV framework is designed to provide query-guided target detection for UAV images and video frames across different aerial conditions. Combining open-vocabulary detection and adaptive processing, the system identifies targets specified through natural-language queries. Overall, the framework consists of sequential stages beginning with query understanding and initial detection, followed by analysis, selection, verification, and output. Specifically, the system includes query understanding, open-vocabulary detection, adaptive detection analysis, SAHI-based processing, Super-Resolution enhancement, detection verification, and final target output.

### A. System Architecture

The proposed system operates through a sequence of processing stages that transforms a UAV image or video frame and a target query into a verified detection result. Initially, the input image or video frame is provided together with a natural-language description of the target. The YOLO-World identifies open-vocabulary based on the given query and detects the relevant target regions. The initial detection results are analyzed using parameters including bounding-box location, confidence score, target size and visual conditions. Based on characteristics, the system determines whether the target can be directly detected or requires further processing. Small or distant targets can be processed using SAHI-based sliced inference, while low-resolution or degraded regions can be enhanced using Super-Resolution. The resulting detections are then reviewed and verified before the final target information is presented. The modular architecture allows the individual processing stages to be managed independently and provides flexibility for adapting the framework to different UAV detection scenarios.

### B. Query Understanding and Target Analysis

The first stage of the AgentUAV framework focuses on understanding the target requested by the user. A natural-language target query is provided along with the UAV image or video frame and is used to guide the initial detection process. YOLO-World processes the query and identifies objects that correspond to the requested target. The resulting detections provide information about the target location, bounding box, confidence score, and approximate target size. These details are then analyzed to understand the difficulty of the detection task. At times when the target is clearly visible and sufficiently large, the initial detection can be used directly. However, at times when the target is small, distant, partially visible, or affected by poor image quality, the system can identify the need for additional processing.

### C. Adaptive Detection Processing

UAV targets can vary greatly in size, distance, resolution, and surrounding visual conditions. Therefore, applying a single processing approach to every input may result in unnecessary computation for simple scenes or insufficient detection for difficult targets. The Adaptive Detection Processing analyzes the initial YOLO-World results and determines the appropriate processing strategy for the detected target. Based on the target characteristics, the system can continue with the standard detection result or activate additional processing through SAHI-based sliced inference or Super-Resolution enhancement. The adaptive strategy allows the framework to provide additional processing only when the initial detection indicates a challenging condition. As a result, the system can address small, distant, and low-resolution targets while reducing unnecessary processing for targets that can already be detected reliably.

### D. Query-Guided Detection and Multi-Agent Processing

AgentUAV combines natural-language query guided detection and multiple specialized processing components. The UAV image or video frame and the target query are initially processed using YOLO-World to obtain the target location, bounding boxes, and confidence information. The detection results are then analyzed to identify conditions that may affect the detection quality. If the target is small or distant, the SAHI Processing Agent can perform sliced inference by dividing the input into smaller overlapping regions. If the target region has insufficient visual information due to low resolution or degradation, the Super-Resolution Agent can enhance the image before further detection processing. The outputs from the different processing stages are then passed to the verification and decision components. This multi-agent arrangement allows the framework to coordinate different detection strategies according to the characteristics of the target rather than applying every processing method to every input.

Query-Guided Detection:

$$D(I, Q) = \text{YOLO} - \text{World}(I, Q)$$

where  $I$  represents the UAV image or video frame,  $Q$  represents the natural-language target query, and  $D(I, Q)$  represents the initial query-guided detection result.

Target Detection Confidence:

$$C = P(T | I, Q)$$

where  $C$  represents the detection confidence,  $T$  represents the target,  $I$  represents the UAV image or video frame, and  $Q$  represents the natural-language target query.

### E. SAHI and Super-Resolution Processing

The SAHI Processing Agent is responsible for handling the targets that are difficult to detect due to their small size or large distance from the UAV. Rather than processing the complete image as a single region, SAHI divides the input into smaller overlapping slices and performs detection on each region. This allows small targets to occupy a larger effective area during inference and can improve their visibility to the detection model. The Super-Resolution Agent focuses on low-resolution or degraded target regions. Real-ESRGAN is used to enhance the visual representation of these regions and recover useful image information before additional detection processing. The two processing methods address different detection challenges and are selected according to the conditions identified during the adaptive analysis stage.

### F. Detection Verification and Adaptive Decision

After the initial and additional detection stages are completed, the obtained results are reviewed by the Detection Verification and Adaptive Decision components. Detection verification considers factors such as confidence score, target location, bounding boxes, and consistency with the original query. The results obtained from standard detection, sliced inference, and Super-Resolution processing are compared to determine whether additional processing has provided useful detection information. The Adaptive Decision Agent coordinates these results and selects the appropriate final detection outcome according to the available evidence. This stage helps reduce unreliable detections and ensures that the final result is based on the combined information produced by the different processing stages.

### G. Final Detection and Output

The final stage prepares the verified target detection results for presentation. After the detection and verification stages are completed, the framework provides the identified target together with its corresponding bounding box and detection information. The final output reflects the original natural-language query and incorporates any additional processing that was selected for the particular UAV scene. In this way, the framework can provide a consistent detection output while adapting the processing pipeline to the difficulty of the target.

### H. End-to-End Workflow

The complete AgentUAV workflow begins with a UAV image or video frame and a natural-language target query. The query is first interpreted and used by YOLO-World to perform open-vocabulary target detection. The initial detection results are then analyzed according to target size, confidence, location, and visual conditions. If the target can be detected directly, the initial result can proceed towards verification. For challenging targets, the system activates appropriate additional processing, such as SAHI-based sliced inference for small or distant objects

or Super-Resolution enhancement for low-resolution regions.

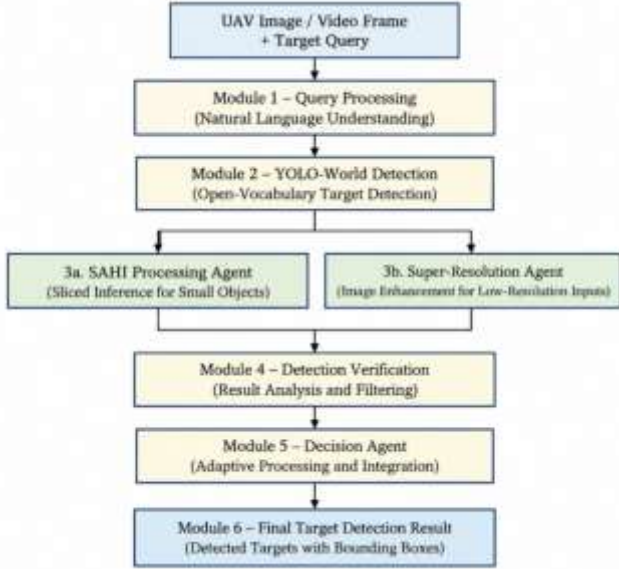


Fig. 1. Architecture of the Proposed AgentUAV Multi-Agent Query-Guided UAV Target Detection Framework

## V. EXPERIMENTAL RESULTS AND EVALUATION

The proposed AgentUAV framework was implemented within the scope of the current research to evaluate the query-guided UAV target detection paradigm for different processing configurations. The framework receives UAV images/video frames with target queries in natural language and performs initial open-vocabulary detection using YOLO-World. It subsequently analyzes the detected results and determines if there are challenging conditions, including small target size, target at a long range, and low image/video resolution. Additional processing steps such as SAHI-based sliced inference processing or Super-Resolution are performed, if needed. The current framework was benchmarked against the baseline YOLO-World detector, YOLO-World + SAHI configuration, and the proposed AgentUAV multi-agent configuration under the same evaluation conditions.

### A. Experimental setup

The experimental evaluation was conducted using the prepared UAV aerial image benchmark and the same target categories for the different configurations. Three detection systems were considered for the comparative analysis. First, a detection system based on standard YOLO-World zero-shot detection was used as a baseline configuration. Second, a configuration based on YOLO-World and SAHI-based sliced inference was considered to determine the impact of smaller image regions processing for challenging targets. Finally, the third configuration represents the complete AgentUAV framework, which includes adaptive planning, SAHI and Real-ESRGAN Super-Resolution processing and detection verification. All configurations were evaluated under the same experimental conditions to ensure that the differences in detection performance were related to additional processing steps.

Table I. Experimental System Configurations

| System | Configuration | Purpose |
|--------|---------------|---------|
|--------|---------------|---------|

|          |                                 |                                                                                                       |
|----------|---------------------------------|-------------------------------------------------------------------------------------------------------|
| System A | Baseline YOLO-World (Zero-Shot) | Baseline detector performance on full UAV aerial frames without slicing or enhancement                |
| System B | YOLO-World + SAHI               | Evaluates the contribution of Slicing Aided Hyper Inference (SAHI) for small and dense aerial targets |
| System C | AgentUAV Multi-Agent System     | Evaluates the complete proposed framework (Adaptive Planning + Real-ESRGAN + SAHI + Verification)     |

### B. Evaluation strategy

To validate the effectiveness of the proposed framework, we compared three configurations of the detector: baseline YOLO-World zero-shot detector, YOLO-World with SAHI configuration, and full AgentUAV multi-agent system. We compared the results of these two approaches to evaluate the contribution of the sliced inference and adaptive multi-stage processing to the improvement of UAV target detection performance. All three configurations used the same images for evaluation and the same set of target categories. The comparison allows us to examine how different processing strategies affect the ability of a model to detect small, distant, and challenging-to-observe objects in an aerial environment.

### C. Evaluation Metrics

The evaluation uses standard object detection metrics to measure the performance of the proposed AgentUAV framework across the different experimental configurations.

#### 1. Precision

Precision measures the proportion of detected targets that are correctly identified.

$$Precision = \frac{TP}{TP + FP}$$

where **TP** represents True Positives and **FP** represents False Positives.

#### 2. Recall

Recall measures the ability of the system to correctly identify all relevant targets present in the UAV images.

$$Recall = \frac{TP}{TP + FN}$$

where **FN** represents False Negatives.

#### 3. F1-Score

F1-Score provides a combined measure of Precision and Recall by calculating their harmonic mean. It is useful for evaluating

the balance between correctly identified targets and missed or incorrectly detected targets.

$$F1 = 2 * \frac{Precision * Recall}{Precision + recall}$$

where *Precision* represents the proportion of correct detections among all predicted detections, and *Recall* represents the proportion of correctly detected targets among all actual targets.

#### 4. Mean Average Precision

Mean Average Precision (mAP) measures overall object detection performance by considering Precision and Recall across different confidence thresholds and object categories.

$$mAP = \left(\frac{1}{N}\right) \sum_{i=1}^N AP_i$$

where *AP* represents the Average Precision for each target category and *N* represents the total number of categories.

#### D. Experimental results

The experimental comparison shows differences between the baseline YOLO-World detector, the YOLO-World with SAHI configuration, and the complete AgentUAV multi-agent framework. The baseline configuration provides the initial zero-shot detection capability, while the SAHI configuration introduces sliced inference to improve the handling of small and distant targets. The complete AgentUAV framework further incorporates adaptive processing, Super-Resolution enhancement, and verification stages.

The baseline YOLO-World configuration achieves a Precision of 29.84%, Recall of 22.31%, and mAP@50 of 7.06%, with an F1-Score of 25.53%. With the introduction of SAHI, Precision increases to 66.50%, Recall reaches 65.38%, and mAP@50 increases to 45.10%, resulting in an F1-Score of 65.94%. The complete AgentSearch-UAV multi-agent configuration achieves a Precision of 61.43%, Recall of 65.03%, and mAP@50 of 45.45%, with an F1-Score of 63.18%. These results indicate that sliced inference and the adaptive multi-stage processing strategy substantially improve detection performance for challenging UAV imagery compared with the baseline zero-shot configuration.

#### E. Discussion

The results of the experiments prove the effectiveness of the proposed adaptive processing for the given task. Compared to the baseline, the YOLO-World architecture allows for faster direct querying of objects; however, its performance significantly decreases when detecting a large number of objects in the given scenario. The addition of SAHI leads to an improvement in the metric, as it allows processing smaller images, thus enabling the model to detect distant and small objects. AgentUAV in its complete configuration, thus, with multi-stage processing and selection of additional candidates based on the obtained results, was able to achieve the best

performance in terms of coverage due to the additional processing steps.

The results also show that depending on the configuration, the characteristics of the system vary significantly. Thus, the configuration with YOLO-World and SAHI achieved the best result in terms of mAP@50 and F1-Score, while the full configuration of AgentUAV achieved the best Recall. The lower value of Precision for the multi-agent is explained by the fact that additional candidates detected in the multi-stage processing steps may have lower confidence and therefore affect the Precision metric. These results confirm the effectiveness of the adaptive approach, as such variations in the characteristics allow for a higher number of targets to be detected; however, the increased coverage also means that the number of false positives may increase, which must be taken into account when using such systems.

Table II: Performance Comparison of the Evaluated Systems

| System                          | Precision | Recall | F1-Score | mAP@50 |
|---------------------------------|-----------|--------|----------|--------|
| Baseline YOLO-World (Zero-Shot) | 29.84%    | 22.31% | 25.53%   | 7.06%  |
| YOLO-World + SAHI               | 66.50%    | 65.38% | 65.94%   | 45.10% |
| AgentUAV Multi-Agent System     | 61.43%    | 65.03% | 63.18%   | 45.45% |

Overall, the results demonstrate that applying the query-guided approach with subsequent adaptive processing is more beneficial for attaining high target coverage with challenging UAV scenes. In particular, SAHI has proven to be more efficient in detecting small objects while the whole AgentUAV pipeline with sliced inference, Super-Resolution, verification, and adaptive decision-making components allows the system to address a variety of detection scenarios with varying levels of difficulty. The experiments were carried out on a small-scale UAV dataset; therefore, it would be more informative to perform additional tests on a larger UAV scene benchmark and apply the methods to additional aerial image sets to evaluate the performance and generalization of the framework further.

#### VI. CONCLUSION AND FUTURE WORK

The proposed AgentUAV framework can be seen as a multi-agent, query-guided system for the detection of targets in UAV imagery under varying aerial conditions. The framework is able to make use of open-vocabulary detection, adaptive processing, SAHI-based sliced inference, Super-Resolution enhancement, and detection verification for the purpose of better processing of challenging targets. To begin with, the framework uses a natural-language query to find the requested target with the help of YOLO-World. Following that, the framework analyzes detection results to see if further processing is needed, where SAHI is used to process small and distant targets while Super-Resolution takes care of low-resolution/destroyed regions.

According to the experimental results, different processing configurations have varying effects on detection performance, with SAHI being beneficial for small-object detection while the entire AgentUAV framework provides greater target coverage using adaptive, multi-stage processing. As a result, the framework provides a versatile solution for query-based UAV target detection in challenging aerial scenes.

The current AgentUAV framework has primarily been developed and evaluated for UAV target detection scenarios involving challenging aerial imagery. Its performance may vary depending on a variety of factors such as target size, image resolution, viewing altitude, object density, environmental conditions, and the availability of suitable target queries. Moreover, a complete quantitative evaluation regarding the SAHI and Full AgentUAV configurations is still limited, which constrains broader conclusions about the comparative performance of these approaches.

The future work should explore the AgentUAV framework using larger and more diverse UAV data sets to further evaluate its generalization and robustness. Additional experiments can be performed to investigate the individual contribution of SAHI, Super-Resolution, detection verification, and adaptive agent-based processing through detailed ablation studies. The framework can be further optimized for real-time UAV applications by reducing the processing requirements and improving the efficiency of the multi-stage pipeline. Future work can also consider supporting additional target categories and evaluating the framework under different environmental, altitude, and surveillance conditions. Such improvements can further enable AgentUAV to be used in more practical UAV-based surveillance, search, monitoring, and target detection applications.

## VIII. ACKNOWLEDGMENT

The authors extend their gratitude to the Intel Unnati Centre of Excellence at Kongu Engineering College, Erode-638060, India for the computing and technical support provided for the completion of their work. We express appreciation for the help and assistance provided by the institution during the entire course of the research work including development and testing.

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